EDUCATIONAL DEMO · SYNTHETIC DATA ONLY

Samvad HealthDesk AI Final Regression & Release-Readiness Report

P0/P1 defect remediation, security hardening and combined-candidate validation

1	11/12	121	16/16	181
P0 FIXED	P1 RESOLVED	MOCK TESTS	E2E CHECKS	BACKEND TESTS

Not release-approved. One P1 remains owner-blocked (real prompt audio), five visual findings remain with the design owner, and this is a disposable validation candidate rather than a release branch.

Report date	3 November 2026
Audit of record	`audit/final-ui-logic-figma-review` @ `ab4ba5b`
Frontend base	`feat/frontend-figma-parity` @ `78b68c2`
OpenCode remediation	`fix/opencode-audit-frontend` @ `c12df34`
Backend baseline	`feat/backend` @ `e5d77e5`
Combined candidate	`validation/samvad-combined-release-audit` @ `938126a`
Disposition	Validation rehearsal only — not merged, not pushed, not deployed

1. Executive summary

The P0 defect and every OpenCode-owned P1 from the 50-item audit were corrected on an isolated branch, then combined with the design owner's visual pass in a disposable worktree and re-tested end to end.

> **All executable suites pass on the combined candidate.** TypeScript is clean, the mock suite is 121/121, the simulated Supabase contract suite is 7/7, headless-Chrome E2E is 16/16, visual QA covers 7 viewports, the production build succeeds with zero fixture or secret strings, and the unchanged backend Deno suite is 181/181.

What changed most: the doctor's warning-flag section now reads the report instead of printing “None”; emergency and confirmation screens can no longer be navigated away from; booking sends a real slot timestamp and renders an Asia/Kolkata date; the daily-cap (`demo_busy`) response is handled instead of silently disabling consent; every fixture bypass and the credential-free doctor path were deleted from shipped code.

What is still open:

- **D-06 / N-02 — placeholder audio.** Eleven files exist in `public/audio`, but all eleven are byte-identical 1-second 8 kHz WAVs. Playback and caching are correct and never call the paid provider; the greeting is still silent. Owner action.
- **Five visual findings** remain with the design owner (processing screen, booking layout, reports list, dedicated slot-state screens, brand plus), plus three partial visual items (review row layout, avatar containers, remaining English copy).
- **Two owner decisions** (urgent-review authority, consent copy) and one process gap (the visual corrections are uncommitted).
- **Real RLS/JWT behaviour** could not be exercised: no local Supabase stack, and hosted access was out of scope.

Recommendation: suitable for an internal scripted demonstration. Not yet for hosted evaluator access or permanent branch integration.

2. Candidate identity

Role	Ref	Commit / identifier
Audit of record	audit/final-ui-logic-figma-review	`ab4ba5b`
Frontend base	feat/frontend-figma-parity	`78b68c2`
Design-owner visual work	feat/frontend-figma-parity	`78b68c2` (src diff sha256 ffd2fd28ce7e45a170e410b0…)
OpenCode remediation	fix/opencode-audit-frontend	`c12df34` (code d7a3248, report 77c9cc5)
Backend baseline	feat/backend	`e5d77e5`
Combined validation candidate	validation/samvad-combined-release-audit	`938126a`

Process note N-03 (resolved): the design corrections were uncommitted when the candidate was assembled and were committed as `78b68c2` during validation. The committed `src/` diff is byte-identical to the snapshot that was merged and tested (sha256 `ffd2fd28ce7e45a170e410b0…`), so the candidate is reproducible from two commits: `938126a` + `78b68c2`.

3. Integration-validation scope

Executed	Not executed
TypeScript · mock unit/integration · simulated Supabase contract · headless-Chrome E2E against the built bundle · visual capture at 7 viewports · production build inspection · bundle secret/fixture scan · backend Deno tests · static RLS/RPC review	Hosted Supabase / Vercel · live Groq, Prisma, Timbre · real RLS and JWT enforcement · physical devices, Safari, Firefox · screen-reader and axe passes · 15-turn cap in the browser (unreachable in mock mode)

4. Original Figma comparison coverage

The audit inspected all **22/22** frames at native resolution, exported 16 vectors and produced 22 comparison sheets. This validation re-uses that baseline and adds fresh captures of the corrected screens. It does **not** claim a new parity score: the five frames the audit rated as material mismatches are still materially different, and the processing screen is still not a real state.

5. Patient UI verification

Behaviour	Expected	Result
Welcome	Language toggle, Start Talking, Type Instead	PASS
Consent	On-screen Yes/No, versioned copy, cached greeting	PASS greeting served from a static asset, no provider call; autoplay blocking is reported honestly with a tap-to-play control
Consent declined	No health data, no report	PASS
Intake	Fixed question order, chips, voice and text	PASS
Transcript	Editable, confirms before accepting	PASS navigates only after acceptance
Review	All allow-listed fields correctable	PASS incl. other symptoms
Submit	Explicit action, then routing and slots	PASS
Booking	Real slots, IST times, book / not now / more dates	PASS
Confirmation	Reference, doctor, real date and time	PASS “Wed, 04 Nov 2026 · 9:00 AM”
Emergency	112 plain text, intake stopped, no booking, no Back	PASS
Hindi	Devanagari throughout	PARTIAL “Turn 0 of 15” keeps an English word

6. Doctor portal verification

Behaviour	Result
Login requires a real session; unmapped users rejected	PASS
No fixture auto-open, even with legacy keys present	PASS enforced by a source-scan test
Dashboard statistics and recent reports from loaded data	PASS
Reports tabs and search	PASS
Report detail renders real fields and real warning flags	PASS
Urgent detail reads the rule from the report row	PASS captured
Fail-closed review authority	PASS
Stored-answer conversation reader	PASS
Appointments: headers match cells, Upcoming default, IST	PASS
Logout returns to login	PASS

7. Button and action verification

The repository control inventory reports 6 audited controls, 0 unlabelled, 0 dead links, 100% accessible names; the audit's 72-control matrix remains the broader reference. Controls re-verified here include start/type-instead, consent yes/no, question chips, send, mic, done, transcript send / speak again / play again, every review edit, submit, don't submit, date chips, more dates, slot chips, book, not now, end demo, doctor login, password reveal, sidebar navigation, report rows, tabs, search, mark reviewed, acknowledge triage and logout.

Previously decorative controls now behave: consent greeting, play question again, try another date, more dates. Controls that previously did nothing now report honestly: consent without a session, slot-load failure, session restore failure.

8. Navigation verification

Transition	Result
Welcome → Consent → Conversation → Review → Booking → Confirmation	forward flow intact
Back from conversation / transcript / review / booking	allowed, state preserved
Back from emergency	**blocked**, no control rendered
Back from confirmations	**blocked**; submitted report renders read-only
Back from session ended / consent declined	**blocked**
Doctor list → detail → back	search and tab preserved
Direct URL /doctor while logged out	login screen only

9. Avatar, icon and asset audit

Adopted from the design pass: Figma sidebar icon paths, assistant and patient line-art vector paths (30×30 / 28×28), edit pencil on review rows, paper-plane send icon and the 92 px confirmation check circle. Still open: the avatar **containers** (D-43/D-44 require removing the white circle), the brand plus metrics (D-45) and a few rounded-square icon treatments. No image or glyph was invented in this validation.

10. Emergency behaviour verification

- Clear trigger stops routine intake, shows the emergency card with plain-text **112** (no `tel:` link) and the translated guidance, and suppresses booking.
- No Back affordance exists on the emergency screen; an E2E assertion enforces it.
- An `urgent_partial` report with the rule id and a null summary reaches the doctor's urgent queue for both mapped doctors.
- General Medicine sees read-only cross-visibility; only the Emergency doctor can acknowledge.

11. Appointment and booking verification

- Confirmation renders a real IST instant (captured); no “Invalid Date”.
- `slot_start` must be an ISO instant; a slot id is rejected with `invalid_slot`.
- One confirmed appointment per report, in both mock and database; the UI reports that specific state instead of “slot taken”.
- Appointment tabs and rows derive from Asia/Kolkata; headers now match their cells.
- Booking hold for flagged reports, emergency ineligibility and the unique-slot constraint are unchanged in the backend.

12. Consent audio verification

Check	Result
Greeting resolved from a cached asset	PASS `/audio/en/greeting.mp3` → fallback `/audio/greeting_en.wav`
Paid provider used for the cached greeting	PASS none; asserted in E2E
Fake 2.5 s “Playing…” timer	PASS removed
Failure reporting	PASS distinguishes “unavailable” from “browser blocked playback”
Adaptive replay caching	PASS one provider call per language+question per page session
Speech actually present in shipped files	FAIL 11 byte-identical 1-second placeholders

13. Provider integration status

Provider	Status
Groq	Not called; server-side only; live model/structured-output check still owner-gated.
Gnani Prisma (STT)	Not called; browser posts WAV to `chat-turn` ; contract unchanged.
Gnani Timbre (TTS)	Not called in validation; fixed prompts never reach it.
Supabase	Not contacted; wiring validated against the simulated suite only.

14. Security and RLS results

Statically verified (unchanged): JWT-derived identity; RLS on sessions, messages and reports; doctor reads scoped through `doctor_profiles` ; draft reports never visible; `SECURITY DEFINER` limited to `book_appointment` , `mark_report_reviewed` and `get_doctor_appointments` with a fixed `search_path` ; emergency rejected by `submit_report` ; one appointment per report.

Dynamically verified: the doctor portal cannot be entered without authentication; no fixture bypass survives in the bundle; the client never writes to protected tables.

Not dynamically verified: real RLS enforcement, JWT expiry, hostile-client behaviour (no local Supabase stack).

> **Correction made during validation (N-01).** The uncommitted visual branch mapped patient names to invented age/gender/concern values in the appointments table and returned a hardcoded date when a slot timestamp was unparseable. Both were rejected at merge so no invented clinical data reaches a clinician.

15. Complete defect register

53 findings: the original 50 (IDs preserved) plus N-01…N-03 discovered during validation. Status reflects the combined candidate.

ID	Sev	Finding	Status	Evidence / note
D-01	P0	Doctor warning flags hardcoded to None	FIXED	Data-driven flags + 11 tests + captured urgent detail
D-02	P1	Back leaves the emergency state and drops 112 guidance	FIXED	Terminal state guard, no Back control, E2E assertion
D-03	P1	Booking confirmation shows Invalid Date	FIXED	ISO slot_start + IST formatter, captured confirmation
D-04	P1	demo_busy at session start is ignored	FIXED	Busy notice on Welcome + disabled consent + retry

ID	Sev	Finding	Status	Evidence / note
D-05	P1	Mic-denied screen not implemented	FIXED	Real denied state, text fallback, retry (Figma layout)
D-06	P1	No cached greeting; every replay is a paid call	OWNER ACTION	Caching/playback fixed; shipped assets are placeholders
D-07	P1	Hindi emergency body stays English	FIXED	t.emergency.guidance; Hindi E2E
D-08	P1	Production fixtures and credential-free doctor path	FIXED	Deleted from src; scan test; bundle scan clean
D-09	P2	Processing screen not a real state	ANTIGRAVITY	Still fixture-only
D-10	P1	Back from confirmation re-opens booking	FIXED	Terminal confirmations; duplicate booking refused
D-11	P1	Back to consent resets the intake pointer	FIXED	Idempotent consent; state restored
D-12	P1	Other symptoms cannot be corrected	FIXED	Row + editor added
D-13	P2	Try another date does nothing	FIXED	Searches the next date with a free slot
D-14	P2	All absences shown as Not provided	FIXED	Six distinct status labels
D-15	P2	Missing information recomputed	FIXED	Reads report.missing_information
D-16	P2	Booking layout differs from Figma	ANTIGRAVITY	Not rebuilt
D-17	P2	Fixture dashboard/list mismatch	FIXED	Single data source; computed badges
D-18	P2	Appointment headers do not match cells	FIXED	Relabelled Department / Reference
D-19	P2	Play question again navigates away	FIXED	Replay handler wired
D-20	P2	No More dates control	FIXED	14-day window control
D-21	P2	Slot times use the browser timezone	FIXED	Explicit Asia/Kolkata formatting
D-22	P2	Slot load error looks like no slots	FIXED	Error state + retry
D-23	P2	Refresh loses the session silently	FIXED	Restore-failure message
D-24	P2	Hidden links are the first tab stops	FIXED	Removed from tab order
D-25	P2	Doctor load errors swallowed	FIXED	Loading/error/retry; last good data kept
D-26	P2	Hardcoded English on patient screens	JOINT / PARTIAL	Hindi labels adopted; some copy remains
D-27	P3	No password reveal	FIXED	Eye toggle added
D-28	P3	No in-flight disabling	FIXED	Submit/Book/Edit disabled while pending
D-29	P2	Reports filter state lost on Back	FIXED	State lifted to the shell
D-30	P2	Urgent review authority ambiguity	OWNER ACTION	Fail-closed: Emergency doctor only
D-31	P2	Review row layout differs	JOINT / PARTIAL	Pencil icon adopted; 3-column layout not rebuilt
D-32	P2	Reviewed-by name hardcoded	FIXED	Resolved from reviewed_by
D-33	P2	Source chip hardcoded	FIXED	Derived from report + input modes
D-34	P2	Consent bubble copy divergence	OWNER ACTION	Decision not recorded
D-35	P2	Urgent detail content hardcoded	FIXED	Uses report data and routing snapshot
D-36	P3	canReview fails open	FIXED	Deny by default
D-37	P2	Submitted report shown as draft after Back	FIXED	Read-only submitted summary
D-38	P3	Language choice not persisted	FIXED	localStorage + documentElement.lang

ID	Sev	Finding	Status	Evidence / note
D-39	P2	Conversation dead end	FIXED	Explicit end-of-session panel
D-40	P3	Silent microphone re-capture	FIXED	Reported instead of re-recording
D-41	P3	Navigate before the answer is accepted	FIXED	Await then navigate
D-42	P3	Appointments default to Today	FIXED	Upcoming default
D-43	P2	Assistant avatar glyph/container	JOINT / PARTIAL	Exact vector adopted; container remains
D-44	P2	Patient avatar glyph/container	JOINT / PARTIAL	Exact vector adopted; container remains
D-45	P3	Brand plus metrics	ANTIGRAVITY	Unchanged
D-46	P3	Gender prints lower-case	FIXED	Capitalised / mapped
D-47	P2	Reports list is a table, not Figma rows	ANTIGRAVITY	Not rebuilt
D-48	P2	No dedicated no-slots/slot-taken screens	ANTIGRAVITY	Inline states only
D-49	P2	Confirmation layout differs	FIXED	Note + 92 px circle adopted
D-50	P3	Severity chip always dark	FIXED	Selection-driven styling
N-01	P1	Uncommitted branch fabricated demographics and a fallback date	FIXED IN CANDIDATE	Rejected at merge; data-driven behaviour retained
N-02	P1	Prompt audio assets are 1-second placeholders	OWNER ACTION	All 11 files byte-identical, no speech
N-03	P2	Design corrections were uncommitted	VERIFIED FIXED	Committed as 78b68c2; src diff byte-identical to the validated snapshot

Totals — P0 1 · P1 12 · P2 29 · P3 11. Fixed or verified **41**, joint/partial **4**, design-owner **5**, owner action **3**.

16. Full automated test results

Suite	Command	Passed	Failed	Skipped	Notes
TypeScript	`npx tsc --noEmit`	—	0	0	strict mode
Mock unit/integration	`npm run test:mock`	121	0	0	19 suites
Supabase contract (simulated)	`npm run test:supabase`	7	0	0	not a real round trip
Browser E2E	`npm run test:e2e`	16/16	0	0	headless Chrome
Visual / QA	`npm run test:visual`	7 viewports	0	0	23 screenshots
Production build	`npm run build:mock`	pass	0	0	111 modules
Backend Deno	`deno test _shared/`	181	0	0	mocked providers
Local Supabase API	`npm run test:api`	0	3 (load)	0	BLOCKED: no local stack

Assertions added for this report: cached-greeting playback from a static asset and no provider call; a real IST booking date/time with no “Invalid Date”; a data-driven warning-flag chip on the urgent detail; no Back control on the emergency screen.

17. Failed, skipped and blocked

- **Blocked:** `npm run test:api` (3 files) — no local Supabase stack; Supabase CLI absent, Docker present but no project provisioned. Hosted access was not substituted.
- **Not run by design:** live provider calls, hosted environments, device/browser matrix, screen-reader and axe passes.
- **No failures** in any executed suite.
- Inherited harness blind spots: `test:visual` asserts document overflow rather than shell clipping, and the control inventory audits a small fixed list. Both are treated as supporting evidence only.

18. Known limitations

1. Prompt audio is placeholder silence (N-02).
2. RLS/JWT behaviour is statically verified only.
3. The mock backend completes in ≤13 turns, so the 15-turn cap UI is not browser-tested.
4. The doctor portal does not restore a session on refresh (refresh returns to login).
5. Visual parity is unchanged for the five frames the audit rated as material mismatches.
6. The candidate is a disposable rehearsal of two branches, not a release branch.
7. Hindi copy improved but incomplete (D-26).

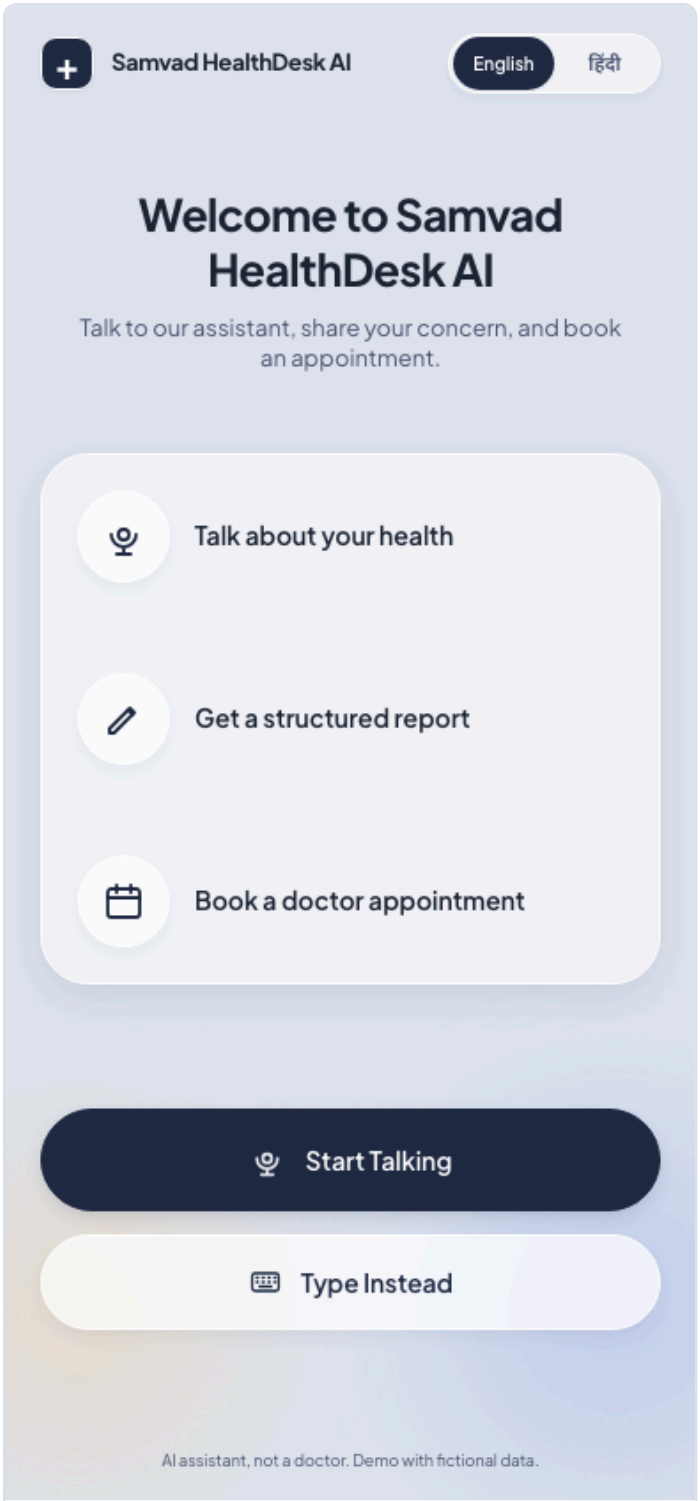
19. Remaining owner actions

1. Generate real prompt audio for the ten documented files (reserve the Timbre budget first); do not ship the placeholders.
2. Record D-30 (urgent-review authority; candidate is fail-closed) and D-34 (consent copy).
3. Rebase the remediation onto `78b68c2` (or re-apply the visual changes on top), resolving the shared screen files per the integration contract.
4. Provision a local Supabase stack and run `npm run test:api`, or accept static security verification explicitly.
5. Decide whether the urgent queue includes `needs_human_review` reports.
6. Approve permanent integration only after 1–4.

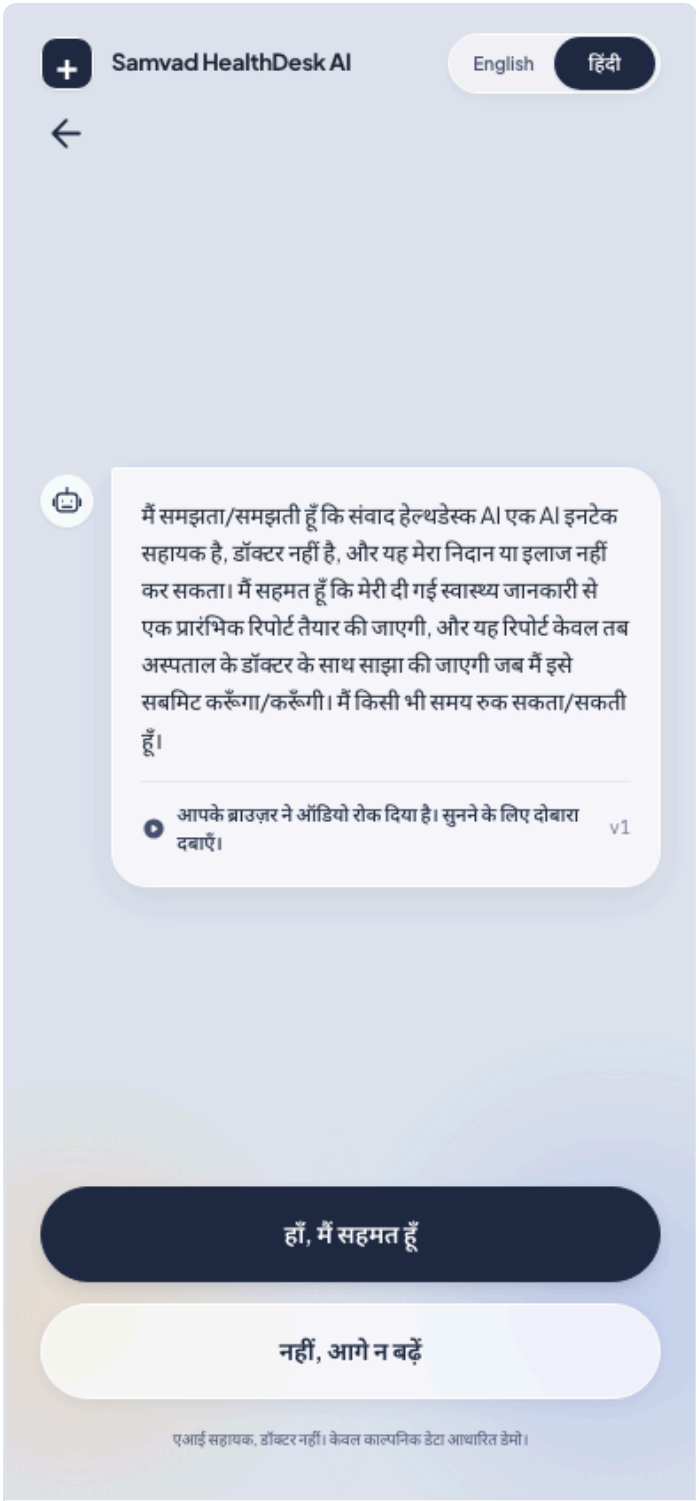
20. Deployability assessment

Gate	Verdict
Internal scripted demo (mock mode)	PASS
Owner visual acceptance	FAIL five visual findings open
Frontend ↔ backend integration	CONDITIONAL contract-tested, not stack-tested
Security review	PASS WITH NOTES
Hosted evaluator access	NOT YET placeholder audio, open decisions
Permanent branch integration	BLOCKED pending owner approval

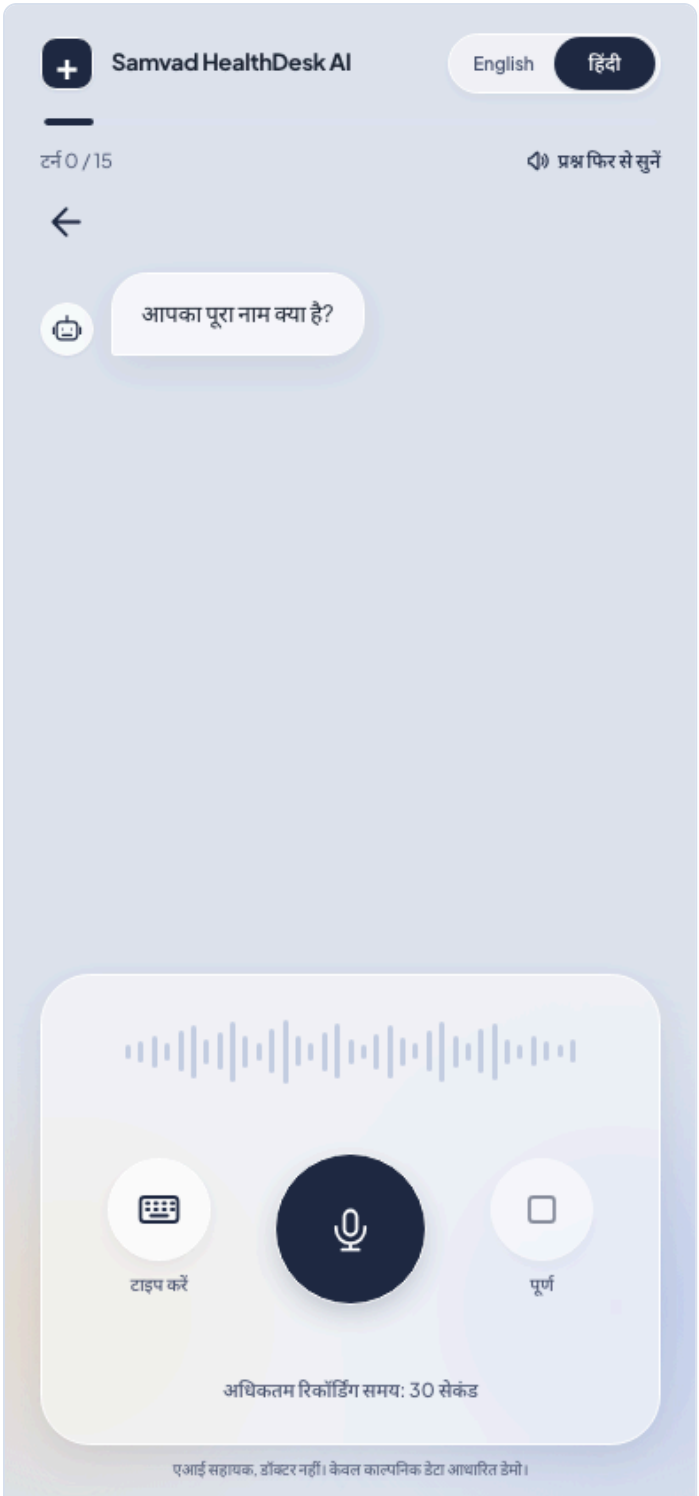
21. Screenshot evidence



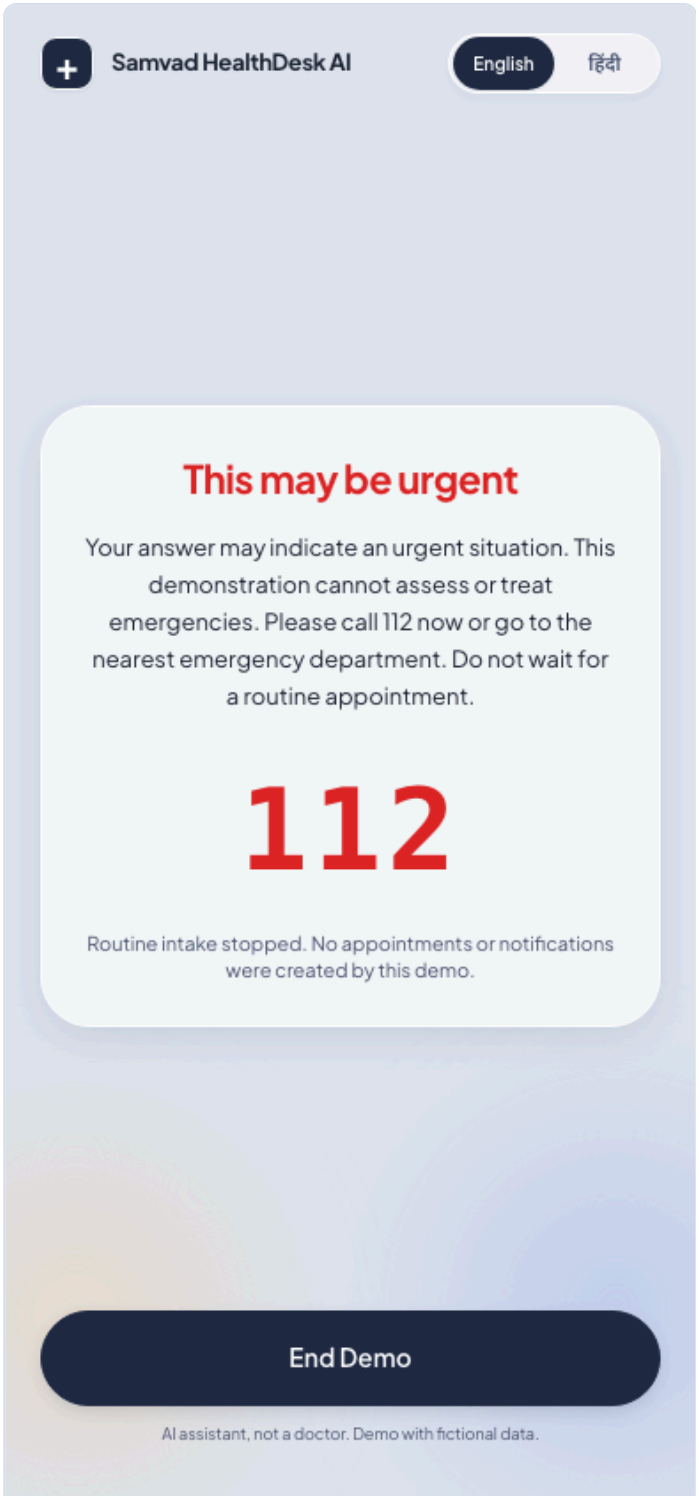
Welcome (mobile)



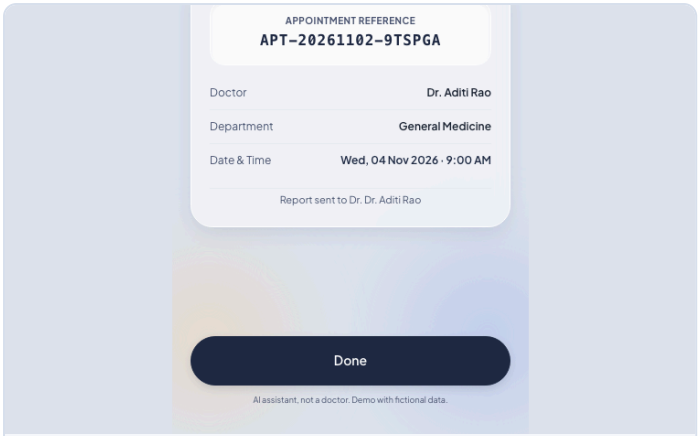
Consent + cached greeting



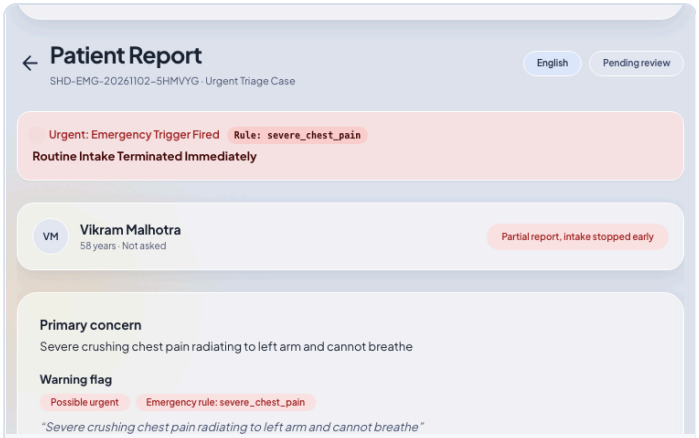
Conversation (Hindi)



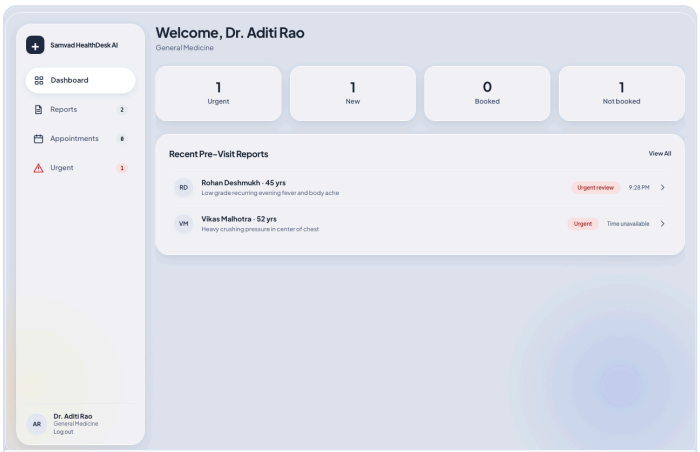
Emergency (112, no Back)



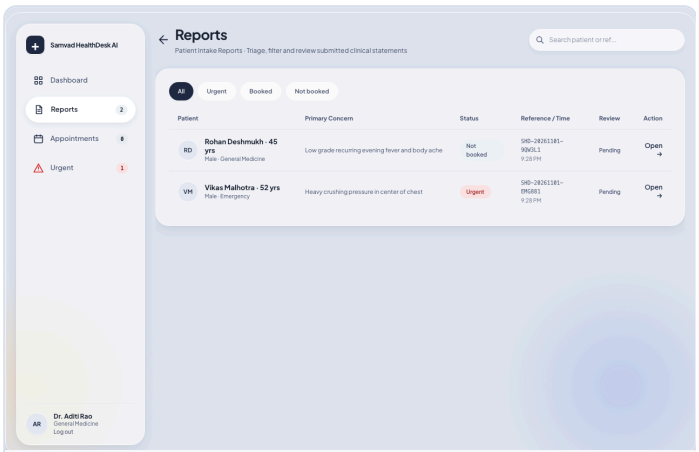
Booking confirmation (IST)



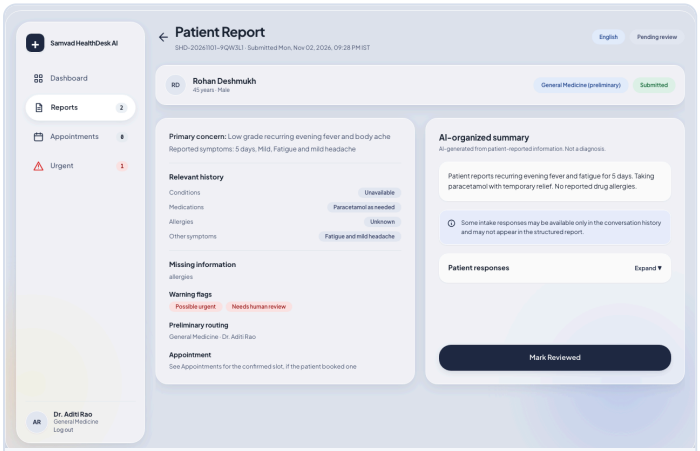
Urgent detail (real flags)



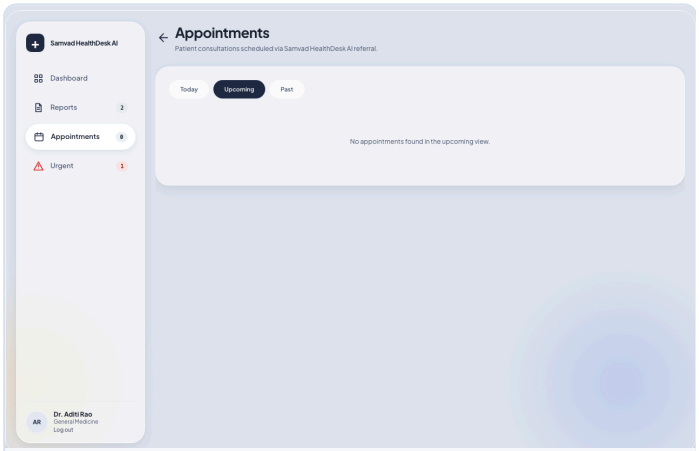
Doctor dashboard



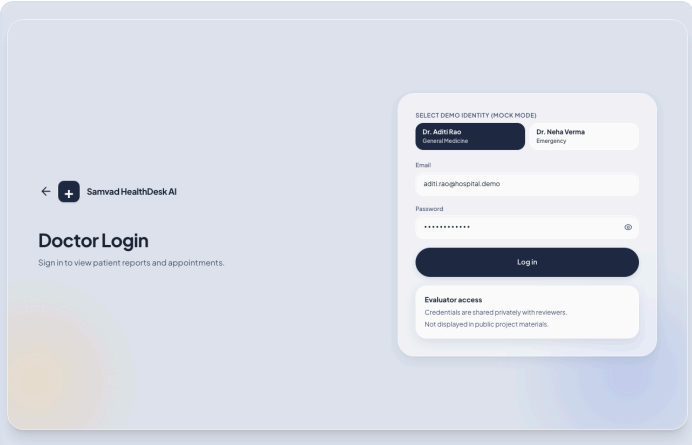
Reports list



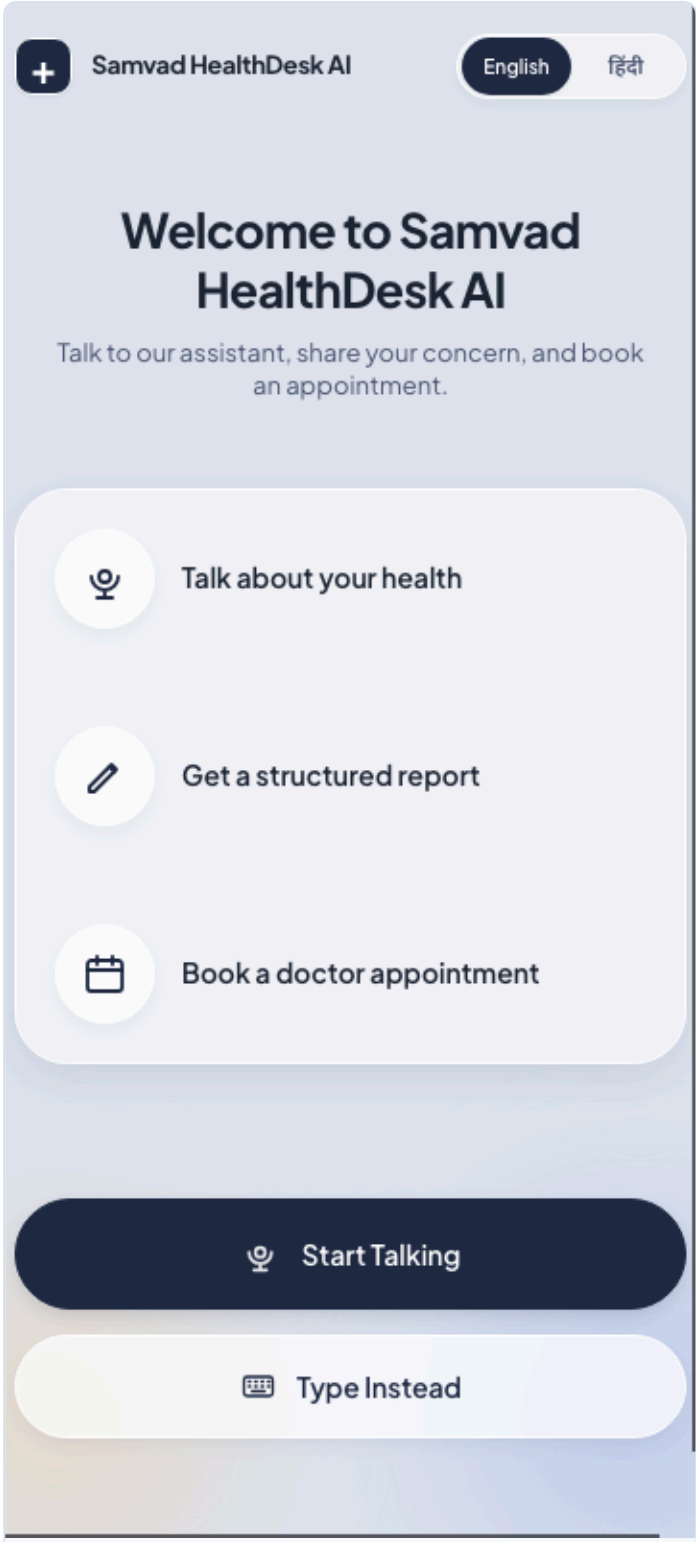
Report detail



Appointments



Doctor login



Welcome (360 px)

22. Evidence index

- Validation worktree: `../samvad-final-release-validation` (branch `validation/samvad-combined-release-audit` @ `938126a`).
- Screenshots: `docs/screenshots/` (23 files).
- OpenCode handoff: `docs/release-remediation/OPENCODE-REMEDIATION-REPORT.md`.
- Integration contract: `docs/release-remediation/OPENCODE-ANTIGRAVITY-INTEGRATION-CONTRACT.md`.
- Audit corpus: `docs/final-audit/` on `audit/final-ui-logic-figma-review`.

No secrets, keys or real patient data appear in this document. All patient and doctor identities used by the application are synthetic demo data.